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# AHMED NOORIM

**B.Sc. in Computer Science and Engineering**

## EDUCATION

January 2025 — **B.Sc. in Computer Science and Engineering**, East Delta University
CGPA: **3.91 out of 4.00**

April 2020 — **Higher Secondary Certificate in Science**, Ispahani Public School & College
GPA: **5.00 out of 5.00**

## RESEARCH INTERESTS

- AI in Healthcare
- Explainable AI
- Computer Vision
- Edge/Cloud Computing
- Medical Image Analysis
- Multimodal Models
- Robotic Vision
- Internet of Things (IoT)
- Biomedical Informatics
- Machine Learning
- Vision in Agriculture
- Human-Computer Interaction

## UNDERGRADUATE THESIS

February 2025 — **An Ensemble Deep Learning Framework for Two-Stage Diabetic Retinopathy Diagnosis and Severity Screening**

- Designed a two-stage diabetic retinopathy (DR) framework: a DenseNet121 detector to detect the existence of DR, followed by severity grading of DR.
- Built a weighted-average ensemble of DenseNet121, Xception, and EfficientNetB1 for four-grade DR severity classification.
- Merged APTOS 2019, MESSIDOR-2, and IDRiD into a 5,922-image dataset and applied augmentation to balance the DR grades.
- Compared three fundus image preprocessing pipelines.

## CONFERENCE PROCEEDINGS

1. **Noorim, A.**, Jahan, R. N., Ahsan, M. S. A., Adhikary, S., & Uddin, M. J. (2025). An ensemble deep learning Framework for Two-Stage Diabetic Retinopathy Diagnosis and Severity Screening. Smart Innovation, Systems and Technologies, 333–342. https://doi.org/10.1007/978-981-95-1365-9_27
2. Uddin, A., Ananna, F. B. F., Farah, N. T., **Noorim, A.**, Hosen, M., Roy, P. S., & Tasnia, N. (2025). PotholeNet11:A YOLO11-Powered Real-Time framework for comprehensive pothole detection and road safety. International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM), 1–6. https://doi.org/10.1109/ncim65934.2025.11160097
3. Farah, N. T., Pushpita, S. H., **Noorim, A.**, Saha, A., Hosen, M., & Uddin, A. (2024). Enhancing Lyme Disease Erythema migrans rashes Classification: U-Net Segmentation and Ensemble Deep Learning Model. IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON), 218–223. https://doi.org/10.1109/becithcon64160.2024.10962629

## POSTER PRESENTATIONS

1. Router-Based Network for Vehicle to Hub Communication to Control Vehicle Speed. *Intra-University Poster Presentation in association with IEEE EDU Student Branch*, East Delta University, 2025. (Poster)

## PROJECT

1. **Vision Based Automated Attendance System.** (Video)
    a. Designed a system that automates QR code scanning and attendance recording for an institution.
    b. Configured an ESP32-CAM with an FTDI module in the Arduino IDE to capture and scan QR codes.
    c. Implemented scan-processing and attendance logic in Python, storing records in MongoDB.
    d. Built a Flask web interface and API layer for the system.

2. **Bengali Handwritten Alphabet Classification Web-App.** (Presentation)
    a. Assembled a dataset of about 20,000 images, combining self-made and internet-sourced images.
    b. Built a CNN model from scratch in TensorFlow to classify single Bengali handwritten alphabets.
    c. Delivered a Flask web interface and API that let users upload an alphabet image and receive its classification.

## WORK EXPERIENCE

June 2024 - Current **Research Assistant**, Research Activator Lab

- Contribute to machine learning research through collaboration and training models.
- Support research writing for lab projects while performing structured metadata extraction.
- Contributed to two IEEE-published papers: *PotholeNet11 (NCIM 2025)* and *Lyme disease rash classification (BECITHCON 2024)*.

March 2018 - Jan 2026 **Graphic Designer**

- *Freelance (Local Market), Mar 2018 – Jan 2026:* Secured and delivered long-term design projects for local businesses using Adobe Photoshop and Illustrator.
- *Marcom Digital Studio, Feb – May 2025:* Designed graphics, illustrations, and branding materials, and gained video editing and color grading skills in DaVinci Resolve.
- *Well Ceramics, Nov 2022 – Mar 2023:* Created brand identity and advertising assets that secured a sales deal worth approximately 1.2 million BDT.

## SKILLS

- **Technical Skills**
    - ***Programming:*** C++, Python, Arduino
    - ***Framework & Tools:*** TensorFlow, pyTorch, Flask, MongoDB, VScode, Kaggle
    - ***Design & Media:*** Adobe Photoshop, Adobe Illustrator, DaVinci Resolve
- **Research Skills**
    - Academic Writing, Boolean Search Queries, Structured Metadata Extraction, Model Fine-tuning
    - ***Models Used:*** U-Net, DenseNet, ResNet, Xception, EfficientNet, YOLO, Transformer
- **Softskills**
    - Leadership and Teamwork
    - Project Management
    - Teaching and Mentoring

## EXTRA CURRICULAR ACTIVITIES

- Vice Convener of East Delta University Erudition Club (May 2022 - Apr 2024)
- Documentation Secretary of ShokherMulluk (April 2018 - Aug 2021)
- Volunteered with Hope Foundation, organizing iftar and distributing Eid gifts to rural children.
- Wrote and performed in the play “Team Chatga,” which won the best performance award at Meraki.
- Ran a food stall at the “Taste and Tunes for Charity” event, generating over 25,000 BDT in sales.
- Led study groups at college and university, arranging concept-clearing sessions for students.
- Captained the CJKS amateur cricket academy team for a month.

## STANDARDIZED ENGLISH TEST

January 2026 **IELTS Overall 7.0** [Listening 8.0, Reading 7.0, Writing 6.5, Speaking 7.0]

October 2025 **GRE Overall 302** [Quant 161, Verbal 141, Writing 3.5]

## HONORS & AWARDS

- Vice Chancellor’s List Award, *East Delta University*
- Dean’s List Award, *East Delta University*
- Board of Trustees Grant Scholarship, *East Delta University*
- Academic Honor “Summa Cum Laude”, *East Delta University*, 2025
- 2nd Runner-Up (Poster Presentation), *East Delta University*, 2025

## REFERENCE

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